

28/11/2023

Midterm

Duration: 90 minutes

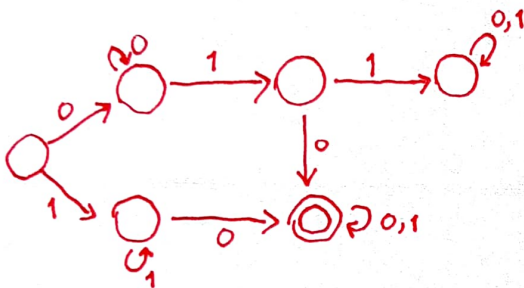
Name:

Solutions

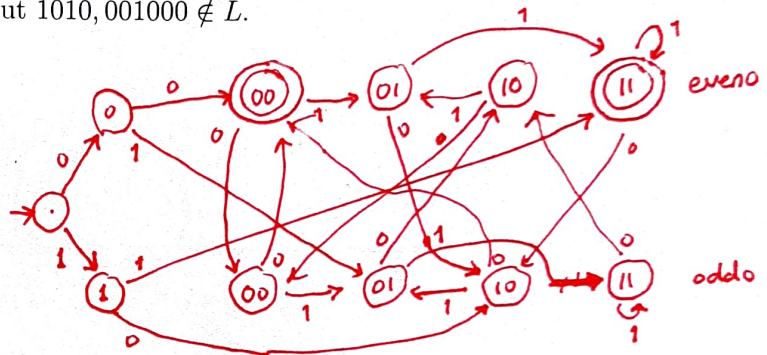
Student No:

P1 [10+15 points] Construct a DFA for the languages below.

(a) $L = \{w \mid w \in \{0,1\}^* \text{ and } 10 \text{ is seen before } 011 \text{ in } w\}$ For example $010, 11011, 101 \in L$ but $001101, 1 \notin L$.

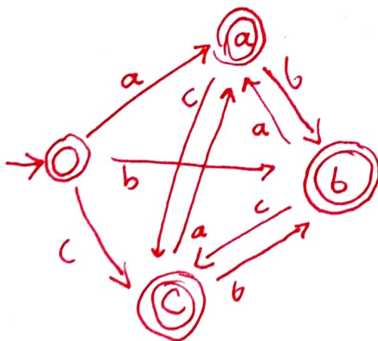


(b) $L = \{w \mid w \in \{0,1\}^* \text{ and } w \text{ has even number of 0s and } w \text{ ends with } 00 \text{ or } 11\}$. For example, $10011, 010100 \in L$ but $1010, 001000 \notin L$.

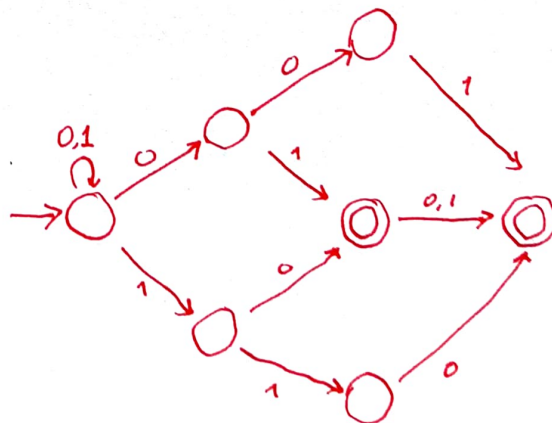


P2 [10+15 points] Construct an NFA for the languages below. Use as few states as you can. Fewest possible gets full credit. Every extra state halves your points.

(a) $L = \{w \mid w \in \{a,b,c\}^* \text{ doesn't contain } aa, bb, cc.\}$ For example, $abc, cbca, \epsilon, b \in L$, but $baac, ccabb \notin L$.



(b) $L = \{w \mid w \in \{0,1\}^* \text{ has at least one 1 and the last three characters contain one or two 0s}\}$. For example, $0110, 11001 \in L$ but $0, 00, 0111, 1000 \notin L$.



[In the questions below, w^R denotes the reverse of w , meaning w written backwards.]

P3 [20 points] (a) Let $\Sigma = \{0, 1\}$ and let $Y = \{ww^Rw \mid w \in \Sigma^*\}$. (For example, $001100001 \in Y$ because it matches the pattern ww^Rw when $w = 001$) Prove that Y is not regular.

Assume that Y is regular. Then, by PL for RL, $\exists p \in \mathbb{Z}^+$ s.t.

$\forall s$ with $|s| \geq p$, s can be splitted as xyz by satisfying PL condition

Let $s = 0^p 1^p 1^p 0^p 0^p 1^p$. To satisfy cond 2&3,

$y = 0^k$ with $k \geq 1$, $x = 0^m$ with $m \geq 0$ and $z = 0^{p-k-m} 1^p 0^p 1^p$.

But then, if we pump down, $s' = 0^{p-k} 1^p 1^p 0^p 0^p 1^p \notin Y$ since

s' doesn't fit into the pattern ww^Rw . Therefore, we have a

contradiction and our assumption must be wrong. Hence, Y is not regular. QED.

1. $\forall i \geq 0, xy^iz \in Y$
 2. $|y| > 0$
 3. $|xy| \leq p$

P4 [15+15 points] Let $L = \{0^n ww^R 1^n \mid n \geq 0, w \in \{0, 1\}^*\}$ For example, $0010000111 \in L$.

(a) Write CFG for L

$S \rightarrow 0S1 \mid E$

$E \rightarrow 0E0 \mid 1E1 \mid \epsilon$

(b) Draw a PDA for L

